Final Exam

● Graded

Student

Huanyu Tang

Total Points

36 / 45 pts

Question 1

Overall Grades **36** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 9 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Huanyu Tang	ht211	Huanyu Tang

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Huanyu Tang

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	5	5	1	0.5	5	36

Problem 1. [5] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

General solution = particular solution + general solution of associated homogeneous system

Associated homogeneous system $BX = 0$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 0 \\ 0 & -2 & -3 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & -2 & -3 & 0 \\ 2 & -3 & 0 & 0 \end{array}\right] \xrightarrow{R_3 = R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & -2 & -3 & 0 \\ 0 & -3 & -4 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 = R_3 - \frac{3}{2}R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & -2 & -3 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{\substack{R_1 = R_1 - 2R_3 \\ R_2 = R_2 + \frac{3}{2}R_3}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_2 = -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$\therefore$ rank $= 3$, no free variables $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ consistent ⇓ (only 1 solution)

Particular solution

① − 2×③ : $-3y - 4z = -1$, $-2y - 3z = 1$ ⇒ $5y = -7z$ ⇒ let $y = 7$, $z = -5$ ⇒ then $x = 11$

So particular solution of $BX = 1$ is $\begin{cases} x = 11 \\ y = 7 \\ z = -5 \end{cases}$

So General solution ⑧ $= \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$

4.5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

row reduction

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow[R_2 = \frac{1}{5}R_2]{R_1 = \frac{1}{2}R_1} \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_4 = R_4 + R_2} \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 5 & 3 \end{bmatrix} \xrightarrow[R_2 \leftrightarrow R_3]{R_4 = R_4 - 2R_3} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

C_1 C_2 C_3

$$\xrightarrow[R_3 = R_3 - 2R_2]{R_3 = R_3 + R_1} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 = R_1 - 2R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(1)

$$\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = (-1)\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + (1)\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$ which is $C_3 = -C_1 + C_2$

So C_1, C_2 are LI (linearly independent), but C_3 redundant

So basis of $Col(A) = \left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) RREF

non-zero row is r_1 and r_2

So basis of $Row(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ -0.5

(3) Let $AX = 0$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

rank = 2, free variable = 1

$x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} s$ let $s = 1$, a basis of null space $= \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

dim(Nullity) = 1

Problem 3. [5] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$e_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}$ $e_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}$ $e_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$ is the standard basis.

$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = 0\,T(e_1) + (1)\,T(e_2) + (-1)\,T(e_3) = T(e_2) - T(e_3) = \begin{bmatrix}1\\1\end{bmatrix}$ ①

$T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = T(e_1) - T(e_3) = \begin{bmatrix}0\\2\end{bmatrix}$ ②

$T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = -T(e_3) = \begin{bmatrix}2\\2\end{bmatrix}$ ③

$\Rightarrow$ So

$$\begin{cases} T(e_1) = \begin{bmatrix}-2\\0\end{bmatrix} \\ T(e_2) = \begin{bmatrix}-1\\-1\end{bmatrix} \\ T(e_3) = \begin{bmatrix}-2\\-2\end{bmatrix} \end{cases}$$

$\because \mathbb{R}^3 \to \mathbb{R}^2 \quad \therefore A$ is a 2×3 matrix

$$\because T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = T(e_1)\cdot x + T(e_2)\cdot y + T(e_3)\cdot z$$

$$= \begin{bmatrix}T(e_1) & T(e_2) & T(e_3)\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}$$

$$= A$$

$$\therefore A = \begin{bmatrix}T(e_1) & T(e_2) & T(e_3)\end{bmatrix} = \begin{bmatrix}-2 & -1 & -2\\ 0 & -1 & -2\end{bmatrix}$$

Problem 4. [5'] Find the Jordan canonical form of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$Av = \lambda v$

$(A-\lambda I)v = 0 \quad \because v \neq 0 \ \therefore$ non-trivial

$\therefore \det(A-\lambda I) = 0$

$$\det \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$$

$= (2-\lambda) \cdot M_{22}$

$= (2-\lambda)(*1)\det \begin{bmatrix} 1-\lambda & 0 \\ -2 & 1-\lambda \end{bmatrix}$

$= (2-\lambda)(1-\lambda)^2$

$= (2-\lambda)^2(1-\lambda)$

$\lambda = 1$

$$A - I = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\text{GM} = 2$, rank $= 2$, So, $\text{GM} = 1$

$\text{AM} - \text{GM} = 2 - 1 = 1$

$$v_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$(A-I)v_2 = v_1 \Rightarrow Av_2 = v_1 + v_2$

$$\Rightarrow \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow v_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$$

$\lambda = 2$

$$A - 2I = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\text{AM} = \text{GM} = 1$

$$v_3 = \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix}$$

So, $A = PJP^{-1}$

$$P = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{2} & 2 \\ 0 & 0 & 1 \\ 1 & 0 & -6 \end{bmatrix}$$

$$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$v = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Jordan blocks $(1, 1, 2)$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

$$f(X) = [x,y,z]^T \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} [x,y,z]$$

Which is $f(x) = x^T A x$

$Av = \lambda v \Rightarrow (A - \lambda 2)v = 0$

$\det(A - \lambda I)$

$$= \det \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$$

$= (3-\lambda) M_{11} + (+1) \cdot M_{13}$

$= (3-\lambda)(+1) \det \begin{bmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{bmatrix} + (+1) \det \begin{bmatrix} 0 & 3-\lambda \\ 1 & -1 \end{bmatrix}$

$= (3-\lambda)(\lambda^2 - 7\lambda + 10)$

$= (3-\lambda)(\lambda-2)(\lambda-5)$

$1^\circ \lambda = 3$ $\quad A - 3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$BM = GM = 1 \Rightarrow \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$

$2^\circ \lambda = 2$ $\quad A - 2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

$BM = GM = 1 \Rightarrow \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$

$3^\circ \lambda = 5$ $\quad A - 5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$

$BM = GM = 1 \Rightarrow \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \Rightarrow v_3 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$

$v_1 \cdot v_2 = 0, \quad v_2 \cdot v_3 = 0, \quad v_1 \cdot v_3 = 0$

$\Rightarrow$ orthogonal. ✓

$\Rightarrow$ normalize ✓

$$P = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$A = PDP^{-1}$ $\quad \because$ symmetric matrix & orthogonal matrix

$x = Pu$ $\quad$ So $A = PDP^T$

$f(x) = x^T A x$

$= (Pu)^T \cdot PDP^T \cdot Pu$

$= u^T \cdot P^T \cdot P \cdot D \cdot P^T \cdot P \cdot u$

$= u^T \cdot I \cdot D \cdot I \cdot u$

$= u^T D u = f(u)$

$f(u) = [u,v,w]^T \cdot \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} \cdot [u,v,w]$

$f(u,v,w) = 3u^2 + 2v^2 + 5w^2$

Problem 6. [5'] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

5K

Let $\begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix} = a\begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} + b\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} + c\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

Const * scalar x Matrix and Matrix + Matrix

$$= \begin{bmatrix} 0 + 2b - c \\ 2a - b + c \\ -2a + 2b + 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} 2b - c = -3 \\ 2a - b + c = 8 \\ -2a + 2b = -8 \end{cases}$$ linear system

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} -2 & 2 & 0 & -8 \\ 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \end{array}\right] \xrightarrow[R_2 = R_2 + R_1]{R_1 = -\frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right] \xrightarrow[R_3 = -\frac{1}{3}R_3]{R_3 = R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -3 \end{array}\right]$$

$$\xrightarrow{R_2 = R_2 + R_3} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow{R_1 = R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

rank = 3 $\Rightarrow$ one solution

$\therefore \begin{cases} a = 3 \\ b = -1 \\ c = 1 \end{cases}$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\} = c_1\begin{bmatrix}0\\1\\1\\0\end{bmatrix} + c_2\begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$ $(c_1, c_2 \in \mathbb{R})$

$\because \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} \neq k\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ So they are Linearly Independent and can span V

$\begin{bmatrix}3\\0\\2\\-1\end{bmatrix} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = 2 \neq 0$ So not orthogonal

So, Let $a, b, m, n \in \mathbb{R}$, $\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = v_1$, $\begin{bmatrix}3\\0\\2\\-1\end{bmatrix} = v_2$

$\begin{cases} n_1 = av_1 + bv_2 \\ n_2 = mv_1 + nv_2 \end{cases}$ ① orthogonal $v_1^2 = 2$, $v_2^2 = 14$

Let $n_1 \cdot n_2 = (av_1 + bv_2)(mv_1 + nv_2)$

$= amv_1^2 + (an + bm)v_1v_2 + bnv_2^2$

$= 2am + 2(an + bm) + 14bn$

$= 0$

$\Rightarrow am + an + bm + 7bn = a(m+n) + b(m+7n) = 0$

$\because n_1 \cdot n_2 = 0$ and $n_1 \neq k n_2$

Find a particular solution

$\begin{cases} a = -4 \\ b = 1 \\ m = 1 \\ n = -1 \end{cases}$

So $n_1 = -4\begin{bmatrix}0\\1\\1\\0\end{bmatrix} + \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} = \begin{bmatrix}3\\-4\\-2\\-1\end{bmatrix}$ $\|n_1\| = \sqrt{30}$

$n_2 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} + \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} = \begin{bmatrix}3\\1\\3\\-1\end{bmatrix}$ $\|n_2\| = \sqrt{20}$

② normalize

$m_1 = \frac{n_1}{\|n_1\|} = \begin{bmatrix}\frac{3}{\sqrt{30}}\\ -\frac{4}{\sqrt{30}}\\ -\frac{2}{\sqrt{30}}\\ -\frac{1}{\sqrt{30}}\end{bmatrix}$

$m_2 = \frac{n_2}{\|n_2\|} = \begin{bmatrix}\frac{3}{\sqrt{20}}\\ \frac{1}{\sqrt{20}}\\ \frac{3}{\sqrt{20}}\\ -\frac{1}{\sqrt{20}}\end{bmatrix}$

① $m_1 \& m_2 \in \mathbb{R}^4$

② $m_1 \cdot m_2 = 0 \Rightarrow m_1 \perp m_2$

③ $\|m_1\| = \|m_2\| = 1$

So m_1 and m_2 are an orthonormal basis for V

0.5/5

Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

$A^T + A = 0$

Since the diagonal numbers don't change, so they are all zero in diagonal ✓

Let $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ $A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}$

$A + A^T = \begin{bmatrix} 2a_{11} & a_{12}+a_{21} & \cdots & a_{1n}+a_{n1} \\ a_{21}+a_{12} & 2a_{22} & \cdots & a_{2n}+a_{n2} \\ \vdots & \vdots & & \vdots \\ a_{n1}+a_{1n} & a_{n2}+a_{2n} & \cdots & 2a_{nn} \end{bmatrix} = 0$

So — ① diagonal number $= 0$, $a_{11} = a_{22} = \cdots = a_{nn} = a_{jj} = 0$

② $a_{ij} + a_{ji} = 0$ with $\Rightarrow a_{ij} = -a_{ji}$ ($i, j \in [1, n]$ and $\in$ positive integer)

$\begin{bmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{bmatrix}$

$\det A = 0 \iff A$ is singular, not invertable

So, because when $\det A \neq 0$, which is A is invertable, n is even. So if n is odd, $\det A = 0$

with $A \cdot A^{-1} = I$

If $n = 3$

$A = \begin{bmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{bmatrix}$

$\det A = -a(-1)\begin{bmatrix} a & -c \\ b & 0 \end{bmatrix} - b(+1)\begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$

$= a(+bc) - b(ac-0)$

$= 0$

So when $n = 2k+1$, $\det(A) = 0$

$\Rightarrow$ This is an example

$n = 2k$, $\det(A) \neq 0$

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2$

$\|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\,\vec{u}\cdot\vec{v} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\vec{u}\cdot\vec{v}$

$\vec{u}\cdot\vec{v} = -\vec{u}\cdot\vec{v}$

$2\vec{u}\cdot\vec{v} = 0$

$\vec{u}\cdot\vec{v} = 0$ So $\vec{u} \perp \vec{v}$

So they are perpendicular to each other